

Lesson 6 : Chap. 3 — classical demand T-tt.



We build demand beh. from scratch, but imposing
fundamental Axioms on consumer preferences and
see what these assumptions imply ab. behavior

The primitive of the model is p.r. $\succeq$.

this hand written into the consumer and captures
evenly about his taste or preference.

If $x, y \in X$ with $x \succeq y$ x is w.p. to y .

From $\succeq$, if $x \succeq y$ but not $y \succeq x$, $x \succ y$.

↓ if
 $x \succeq y$ and $y \succeq x$, then consumer is $\sim$, $x \sim y$.

so we build $\succ$ and $\sim$ from $\succeq$, the latter is the primitive

We start with axioms imply that the consumer's pref. are relations in basic sense.

- completeness: $x, y \in X$, either $x \succeq y$ or $y \succeq x$, or both.
- transitivity: if $x \succeq y$, $y \succeq z$, then $x \succeq z$

This just tells us that the cons. is rational, but it does not tell us about what his tastes are like.

the remain, axioms has more econ meaning.

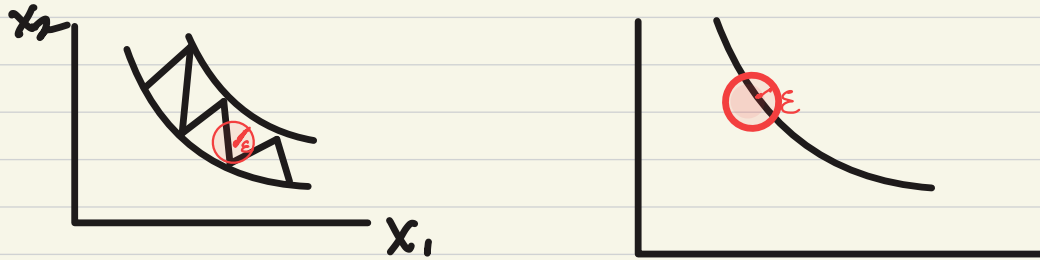
Local non-satiation: For event $x \in X$ and any $\varepsilon > 0$
 $\exists y \in X$ s.t. $\|x - y\| < \varepsilon$ and $y \succ x$.

$\Rightarrow$ There's no bliss point, the consumer

is never SATISFIED because he can always a better bundle close by.

Two important implications

- indifference curves are thin - kill the thick zone.



- The consumer optimize all spends his entire budget constraint bundle.

By contradiction, the budget is not exhausted.

, then it's feasible to change the bundle a bit, mean. that you can find something better.

This contradicts the fact that the original bundle is optimal.

Axioms with more content rehandling the structure of preferences

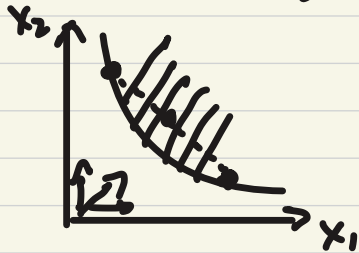
- Monotonicity $\exists$ is monotone if $x \gg y$.
implies $x \succ y$.

$x \succeq y$ is strongly monotone if $x \geq y$ and $x \neq y$, then $x \succ y$.

Monotonicity implies local non-satiation because consumer would always like more of everything.

- convexity $\succeq$ is convex if $\forall z \succ x$ and $\forall z \succeq z$ and $z \succeq x$ implies $\alpha y + (1-\alpha)z \succeq x$.

AVR. always better than extreme.



Similarly, superior set is a convex set $\{y \in \mathbb{R}^2 \mid y \succeq x\}$

strict convexity $\succeq$ is strict $\succeq$ is strictly convex
if $y \succeq x, y \succeq z, z \succeq x$ implies $\alpha y + (1-\alpha)z \succ x \forall \alpha \in (0,1)$
The preference relation, $\succeq$, is abstract. to may become
use to thinking about utility functions. A utility func.
 $U: X \rightarrow \mathbb{R}$ with $x \succeq y \Leftrightarrow U(x) \geq U(y)$

The utility function is just a made-up construct
that is invented to represent preferences in a
diff. way.

Utility functions are of TEN more convenient
to work with, especially if we add assumptions s.a.
differentiability.

However, not all $\succeq$ can be represented at a utility function.

The most famous example, LEXICographic preference.

put the bundle with the most x_2 .

→ to use utility func., more assumption is needed

At least, the extra assumptions has a natural econ interpretation. We need continuity.

$\succeq$ is continuous. If for any sequence

$$\{(x^n, y^n)\}_{n=1}^{\infty} \text{ with } x^n \succeq y^n$$

$\forall N$ and $x = \lim_{n \rightarrow \infty} x^n, y = \lim_{n \rightarrow \infty} y^n$, it holds $x \succeq y$

No sudden preference reversal.